An Explicit Finite-Time Three-Vortex Collapse: A Disproof of Conjecture 00000005268

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Abstract

Three nonzero, mixed-sign circulations admit an explicit solution of the planar point-vortex equations whose vortices remain pairwise distinct until time 1/2 and converge to a common point at that time. The construction uses rational initial coordinates and a logarithmically rotating, square-root shrinking similarity factor. Lean verifies the actual complex-valued differential equation, including its conventional $1/(2\pi)$ factor, and the full one-sided collision limits.

Source and reading. Both source languages assert that finite-time collision of three point vortices is impossible for general strength triples. We interpret "general" as imposing no same-sign restriction. A solution that is collision-free on $[0, T)$ and has a common finite limit as $t \uparrow T$ is a finite-time total collision. The equation need not remain defined at the collision itself. One such solution disproves the universal exclusion, regardless of the proposed additional first integral. This does not assert that collisions occur for same-sign strengths.

Equation and circulations. Identify the plane with $\mathbb{C}$ and use the usual convention

$$\dot{z}_i = \frac{1}{2\pi} \sum_{j \neq i} K_j \frac{\mathrm{i}(z_i - z_j)}{|z_i - z_j|^2}, \qquad i = 0, 1, 2. \tag{1}$$

Set $K_j = 2\pi\gamma_j$, where

$$(\gamma_0, \gamma_1, \gamma_2) = \left(-\frac{1073}{84}, -\frac{1073}{140}, \frac{1073}{224} \right).$$

All three physical circulations are nonzero. With this choice, (1) is precisely $\dot{z} = F(z)$, where

$$F_i(z) = \sum_{j \neq i} \gamma_j \frac{\mathrm{i}(z_i - z_j)}{|z_i - z_j|^2}.$$

This passage is a change of circulation units, not a change in the physical equation or the meaning of collision.

A rational similarity configuration. Define

$$c = \frac{3}{8} - \frac{15}{56}\mathrm{i}, \qquad (w_0, w_1, w_2) = \left(-\frac{3}{8} + \frac{15}{56}\mathrm{i}, \frac{5}{8} + \frac{15}{56}\mathrm{i}, \frac{8}{7}\mathrm{i} \right), \qquad \lambda = -1 - \frac{238}{15}\mathrm{i}.$$

The squared pairwise distances of w are respectively $1, 29/32, 37/32$, so the points are distinct. Direct rational arithmetic in the displayed definition of F gives

i	$F_i(w)$	λw_i
0	$37/8 + (1591/280)\mathrm{i}$	$37/8 + (1591/280)\mathrm{i}$
1	$29/8 - (1711/168)\mathrm{i}$	$29/8 - (1711/168)\mathrm{i}$
2	$272/15 - (8/7)\mathrm{i}$	$272/15 - (8/7)\mathrm{i}$

Thus $F_i(w) = \lambda w_i$. For any $u \neq 0$ and any $c \in \mathbb{C}$, the actual vortex field satisfies the similarity identity

$$F_i(c + uw) = \frac{u}{|u|^2} F_i(w). \tag{2}$$

Indeed, each difference acquires a factor u, and its squared modulus acquires the factor $|u|^2$.

Theorem 1. *The following solution of* (1) *undergoes a total collision at time* $T = 1/2$*:*

$$q = \frac{1}{2} + \frac{119}{15}\mathrm{i}, \qquad u(t) = \exp\bigl(q \log(1 - 2t)\bigr), \qquad z_i(t) = c + u(t) w_i \quad (0 \le t < 1/2).$$

Its initial positions are 0, 1, *and* $3/8 + (7/8)\mathrm{i}$.

Proof. For $t < 1/2$, let $s = 1 - 2t > 0$. Complex exponentials never vanish, and

$$|u(t)|^2 = \exp(2 \operatorname{Re}(q) \log s) = s, \qquad u'(t) = \frac{-2q}{s} u(t) = \frac{\lambda}{s} u(t).$$

Consequently the three positions remain distinct. Differentiation and (2) give, for each i,

$$\dot{z}_i(t) = \frac{\lambda u(t)}{s} w_i = \frac{u(t)}{|u(t)|^2} F_i(w) = F_i(z(t)),$$

which is exactly (1) for $K_j = 2\pi\gamma_j$. Since $u(0) = 1$, adding c to the listed w_i gives the initial positions in the theorem. Finally $|u(t)| = \sqrt{1 - 2t} \to 0$ as $t \uparrow 1/2$. Hence $z_i(t) \to c$ for all three vortices. □

Formal verification and scope. The Lean file defines the vector field by an actual sum over `Fin 3`, using Mathlib complex numbers and their squared norm. It proves the rational field identity, similarity covariance, nonzero circulations, exact initial positions, injectivity of the configuration before collision, and the actual derivative statements `HasDerivAt`. The collision is a Mathlib filter limit from the left. The final theorem negates a universally quantified proposition excluding any such solution for all nonzero circulation triples, all positive finite times, all trajectories, and all collision points. The conventional $1/(2\pi)$ equation and its circulation rescaling are also proved in Lean. No numerical simulation, unverified ODE existence assertion, or endpoint substitution replaces the trajectory.

The separate Hamiltonian formulation and general classification of collapsing triples are not formalized or needed here. The paper's direct proof and the Lean theorem concern this explicit solution. The axiom audit permits only `propext`, `Classical.choice`, and `Quot.sound`.

Source and reproduction. The exact bilingual source is preserved as `SOURCE.md`; the frozen upstream commit and source digest are recorded in `verification/SOURCE_PROVENANCE.md`. The formal proof is in `lean/Main.lean`, using Lean 4.19.0 and the pinned Mathlib dependency recorded in the project manifest. From the Lean project directory, run

```
lake env lean -DwarningAsError=true Main.lean
```

The accompanying strict log contains the actual compiler result and axiom output. Development was by a delegated AI agent with a self-review; the coordinating agent reviews it separately. No external independent review is claimed.